

Sitara Project B'd (Rev.A)

[1] Index

Rev	Date	Designer	Description
A	25.12.20	Ganghyeok Lim	Create design project

Index

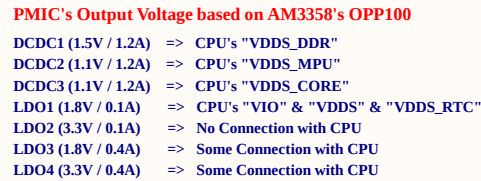
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- #5 Sitara AUX
- #6 Sitara DDR
- #7 Sitara I/O Bank 1/2/3

#8 Sitara I/O Bank 4/5
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#10 Sitara Connectors

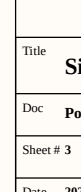
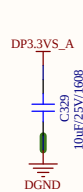
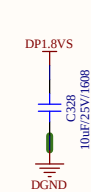
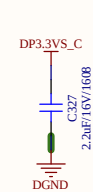
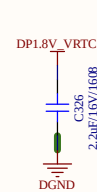
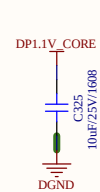
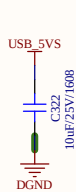
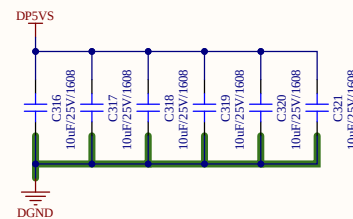
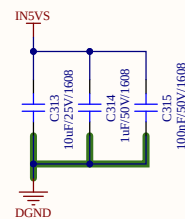
Title		Sitara_Project_Rev_A.PrjPcb		Rev	*
Doc		Index.SchDoc			
Sheet # 1 of 10		Author *			
Date		2025-12-28			
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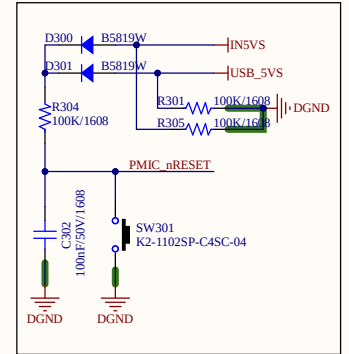
	1	2	3	4																								
A	Sitara Project B'd (Rev.A) [2] Overview																											
B																												
C																												
D	<table><tr><td colspan="3">Title</td><td>Rev</td></tr><tr><td colspan="3">Sitara_Project_Rev_A.PrjPcb</td><td>*</td></tr><tr><td colspan="2">Doc</td><td colspan="2">Overview.SchDoc</td></tr><tr><td>Sheet # 2</td><td>of 10</td><td>Author</td><td>*</td></tr><tr><td colspan="2">Date</td><td colspan="2">2025-12-28</td></tr><tr><td colspan="4">This document is copyright of ArkX and shall not be revealed, produced, copied, in whole or in part, nor used for any purpose other than submitted.</td></tr></table>				Title			Rev	Sitara_Project_Rev_A.PrjPcb			*	Doc		Overview.SchDoc		Sheet # 2	of 10	Author	*	Date		2025-12-28		This document is copyright of ArkX and shall not be revealed, produced, copied, in whole or in part, nor used for any purpose other than submitted.			
Title			Rev																									
Sitara_Project_Rev_A.PrjPcb			*																									
Doc		Overview.SchDoc																										
Sheet # 2	of 10	Author	*																									
Date		2025-12-28																										
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	1	2	3	4																								



Decoupling Capacitors



PMIC has Internal Pull-up Resistor

[illegible]

PMIC_INT_LDO	C303	100nF/50V/1608	DGND
PMIC_BYPASS	C304	10uF/25V/1608	DGND

PMIC_I2C_SCL R320 1.5K/2012
PMIC_I2C_SDA R321 1.5K/2012

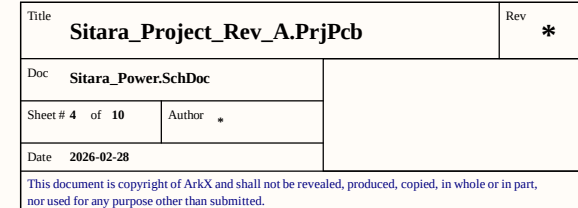
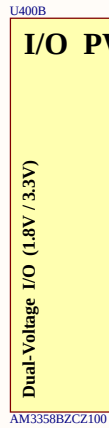
PMIC_nWAKEUP R327 4.7K/1005 DP1.8V_VRTC
PMIC_nINT R328 4.7K/1005 DP3.3VS_A

DP3.3VS_C — LED303 — GREEN/2012 — R334 (510R/1608) — DGND

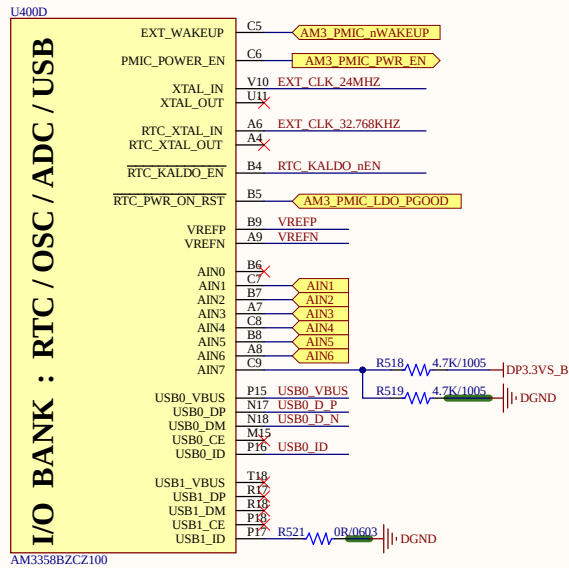
Rail	Output [V]	Sequence (Strobe)
DCDC1	1.5 [V] / 1.2 [A]	1
DCDC2	1.1 [V] / 1.2 [A]	5
DCDC3	1.1 [V] / 1.2 [A]	5
LDO1	1.8 [V] / 0.1 [A]	15
LDO2	3.3 [V] / 0.1 [A]	3
LS1 or LDO3	1.8 [V] / 0.4 [A]	2
LS2 or LDO4	3.3 [V] / 0.4 [A]	4

Place Decaps Closely to Each Pin of PMIC

Title	Sitara_Project_Rev_A.PrjPcb		Rev	*
Doc		Power.SchDoc		
Sheet # 3 of 10		Author *		
Date		2026-02-05		
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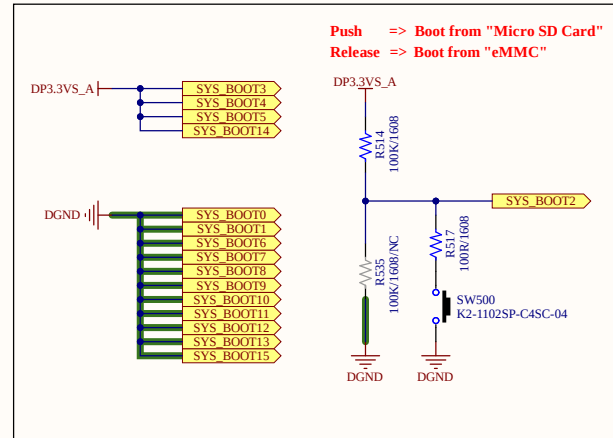
[5] Sitara AUX Part



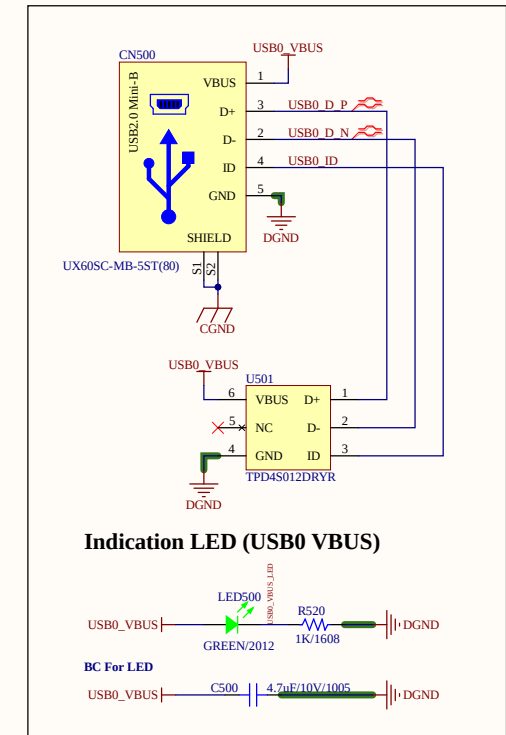
USB Power



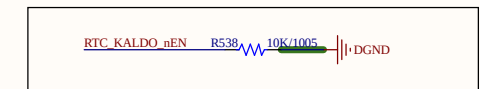
Boot Config



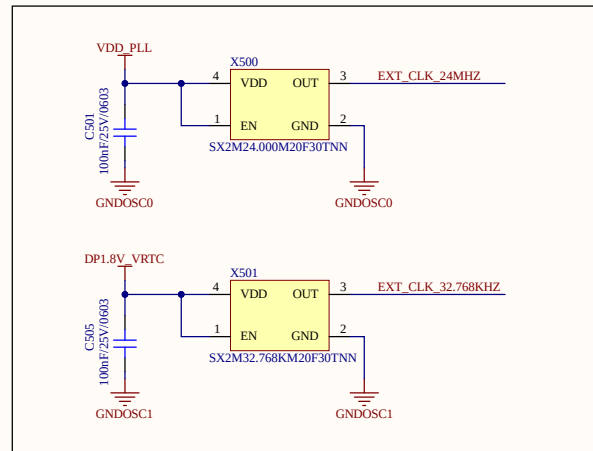
USB 2.0 Mini-B Connector



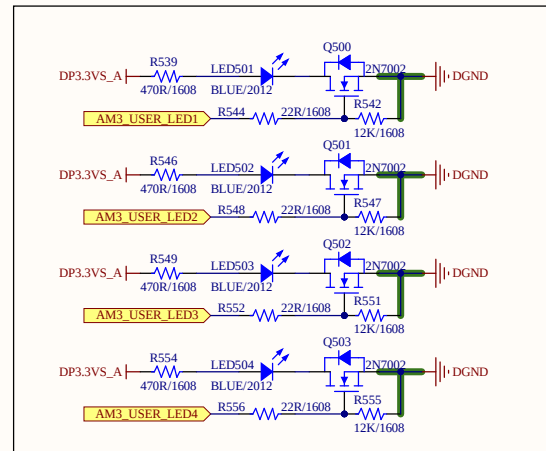
Internal RTC LDO Enable



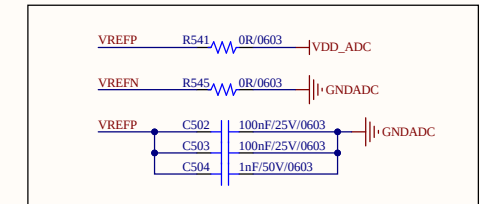
External Oscillator (24MHz / 32.768kHz)



User LEDs



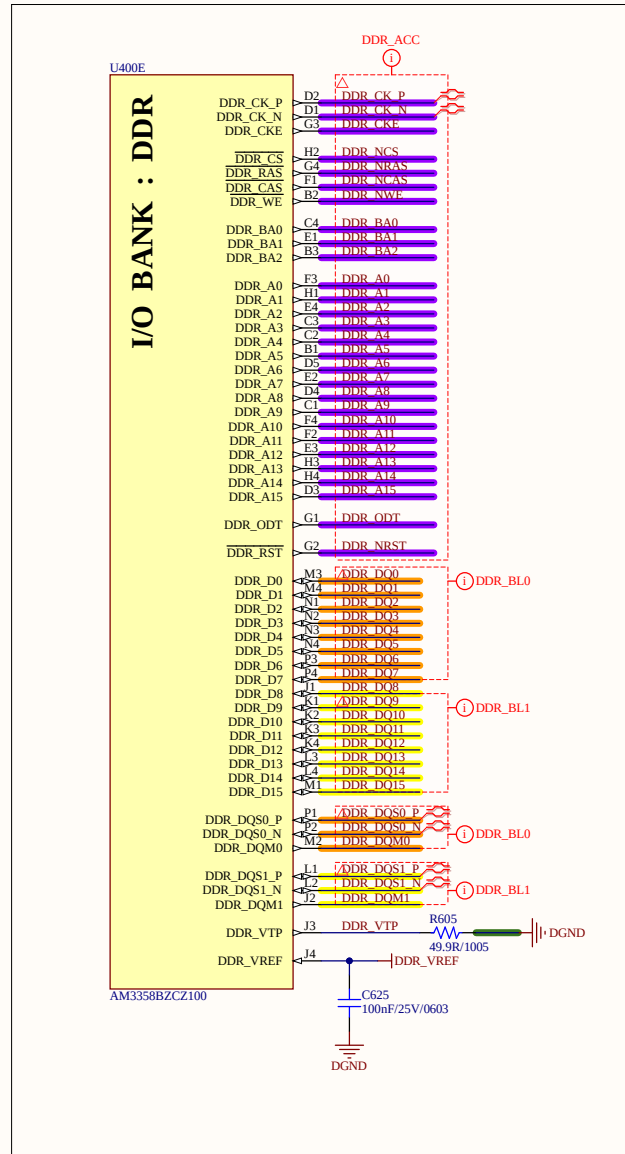
Analog Reference



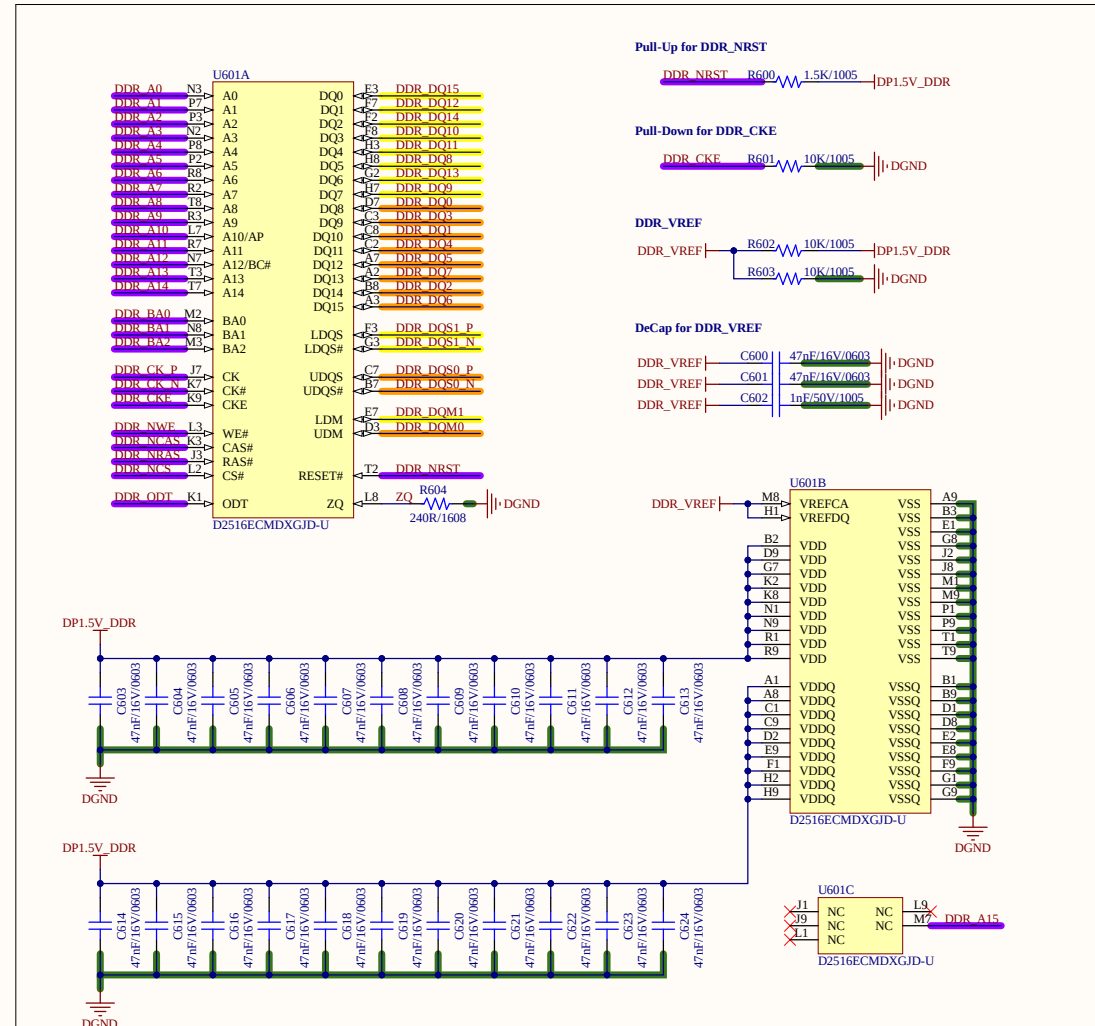
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Doc	Sitara_AUX.SchDoc			
Sheet # 5 of 10	Author	*		
Date	2026-03-02			
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[6] Sitara DDR Interface

Sitara DDR Memory Controller Interface (EMIF)



DDR3L Memory



Title Sitara_Project_Rev_A.PrjPcb		Rev *
Doc Sitara_DDR.SchDoc		
Sheet # 6 of 10	Author *	
Date 2026-02-14		
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Place Serial Resistors Close to MPU

DP3.3V_S_B

MMC0_CLK R710 22R/1005

MMC0_CMD R712 22R/1005

MMC0_DAT0 R713 22R/1005

MMC0_DAT1 R714 22R/1005

MMC0_DAT2 R715 22R/1005

MMC0_DAT3 R716 22R/1005

MMC0_AM3

MMC0_MICRO_SD

DP3.3V_S_B

DP3.3V_S_B

Micro SD

CN700

VDD

CLK

CMD

DAT0

DAT1

DAT2

DAT3/CD

VSS

TF-01A

DGND

CGND

DP3.3V_S_B

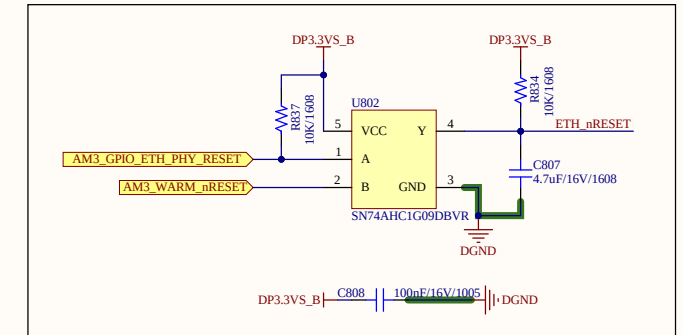
C701 10uF/25V/1608

C702 100nF/16V/1005

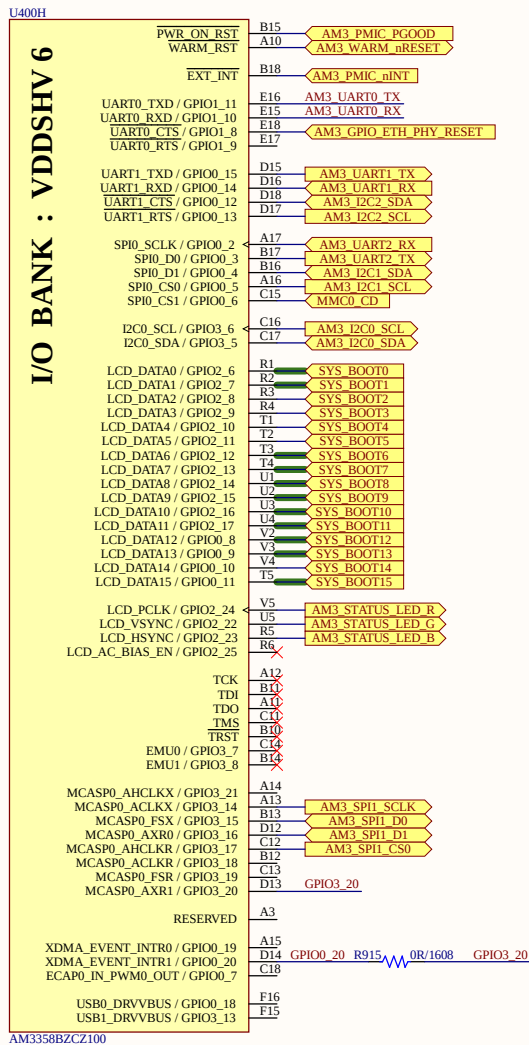
DGND

DGND

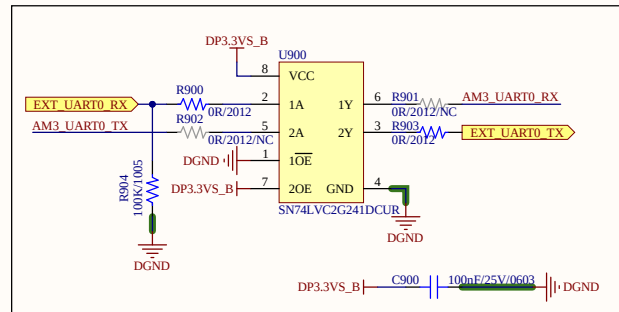
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Doc Sitara_IO_Bank_123.SchDoc		
Sheet # 7 of 10	Author .	
Date 2026-03-01		
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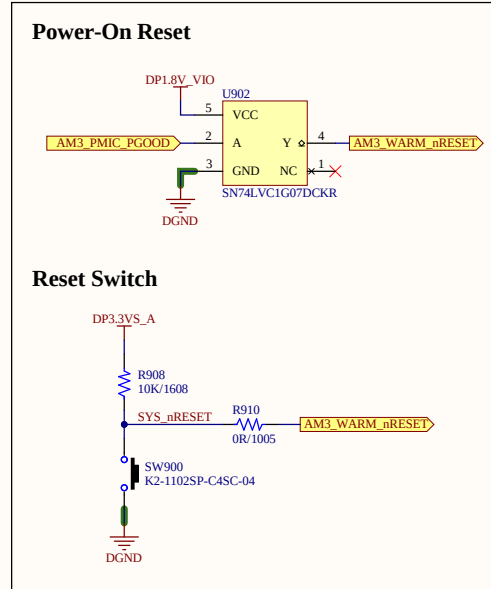
[9] Sitara I/O Bank 6



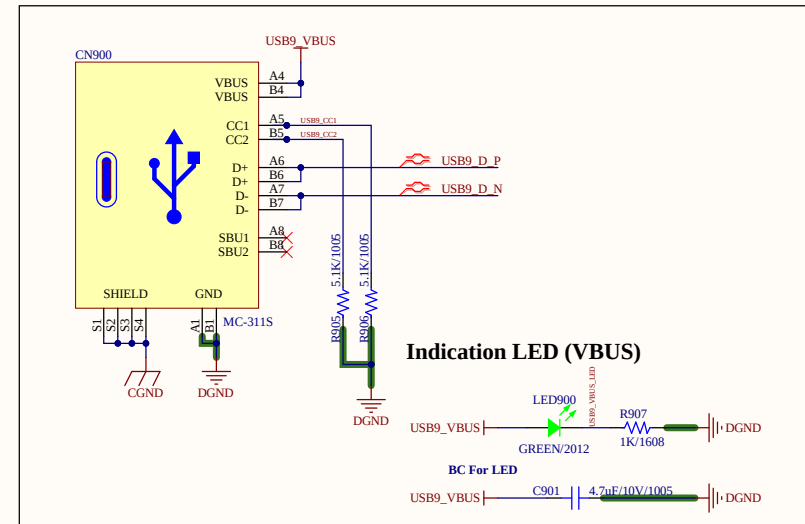
UART0 External Connector



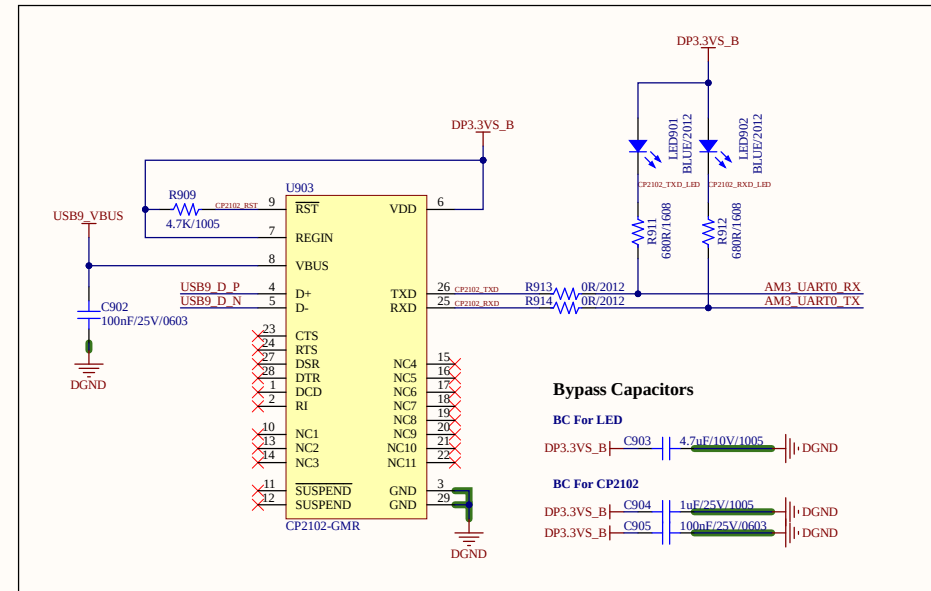
System Reset



USB Connector (Type-C)

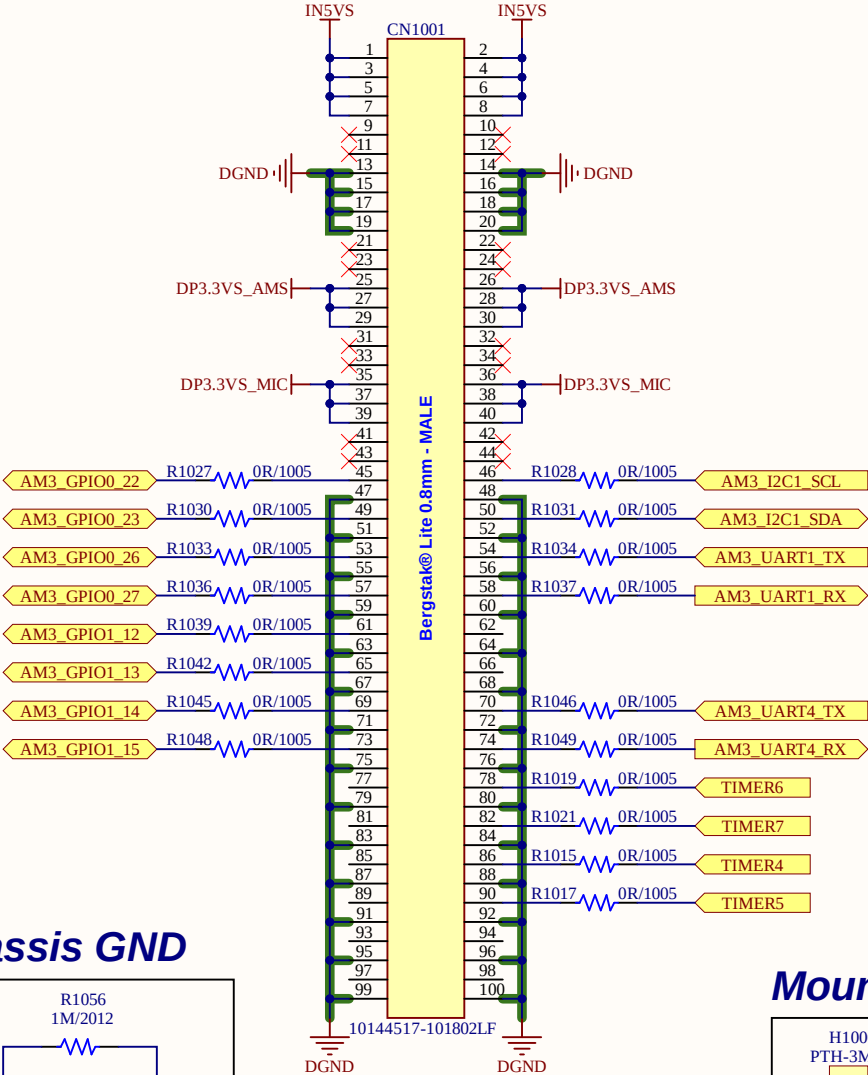


USB - UART Bridge

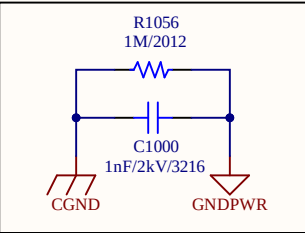


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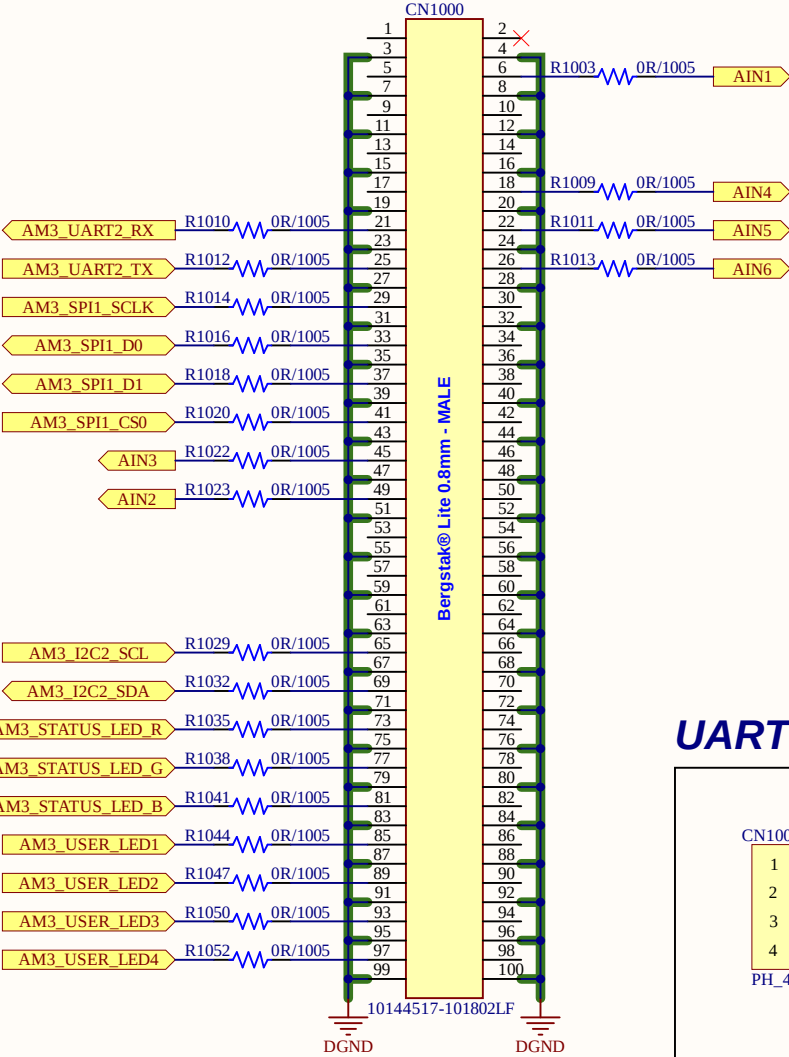
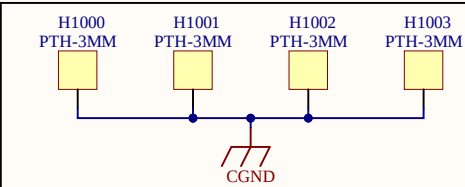
[10] Connectors



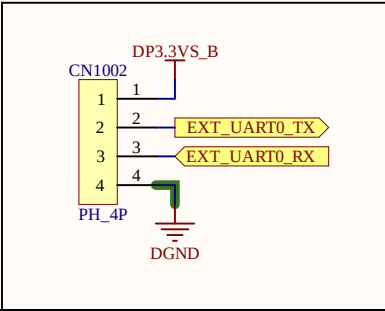
Chassis GND



Mount Holes



UART Connector



Title		Sitara_Project_Rev_A.PrjPcb		Rev	*
Doc		Sitara_Connectors.SchDoc			
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